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STORAGE APPARATUS FOR STORING SHEET

Abstract

A cassette for storing a sheet to be fed includes a first tray portion configured to support the sheet, a second tray portion configured to support the sheet, a connection portion for connecting the first tray portion and the second tray portion, and a grip portion or grip portions provided on the connection portion and used when a user holds the cassette.

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Background/Summary

BACKGROUND OF THE DISCLOSURE
Field of the Disclosure

[0001] The present disclosure relates to a storage apparatus for storing sheets.
Description of the Related Art
[0002] In image forming apparatuses such as copying machines, printers, and facsimile machines, sheets are stored in sheet feed cassettes. Image forming apparatuses that can treat long sheets, not only common office standard-sized sheets such as A4, are widely known.
[0003] As a method of storing the long sheets, Japanese Patent Application Laid-open No. 2021-17335 discusses a sheet feed cassette that can be extended and connected, to treat the long sheets.
SUMMARY OF THE DISCLOSURE
[0004] The present disclosure is directed to a cassette with a high workability.
[0005] According to an aspect of the present disclosure, a storage apparatus includes an apparatus main body, and a cassette for storing a sheet to be fed and configured to be pulled out from the apparatus main body and demounted from the apparatus main body. The cassette includes a first tray portion configured to support the sheet, a second tray portion configured to support the sheet, a connection portion configured to connect the first tray portion and the second tray portion, a first grip portion used when the cassette is pulled out from the apparatus main body, and a second grip portion provided on the connection portion and used when the cassette is held in a state where the cassette is demounted from the apparatus main body.
[0006] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is a cross-section diagram illustrating a color laser printer (image forming apparatus) to which the present disclosure is applicable.
[0008] FIG. **2** is a perspective diagram illustrating a sheet feed cassette to which the present disclosure is applicable.
[0009] FIG. **3** is a perspective diagram illustrating a state where the sheet feed cassette is slid and extended.
[0010] FIG. **4** is a perspective diagram illustrating a state where the sheet feed cassette is separated.
[0011] FIG. **5** is a perspective diagram illustrating a state where the sheet feed cassette is separated when a third tray portion is to be added to the sheet feed cassette.
[0012] FIG. **6** is a perspective diagram illustrating the third tray portion.
[0013] FIG. **7** is a perspective diagram illustrating a state where the third tray portion is additionally connected.
[0014] FIG. **8** is a perspective diagram illustrating a state where the third tray portion is additionally connected and a second grip portion is open.
[0015] FIG. **9** is a perspective diagram illustrating a state where a third tray portion is additionally connected, according to a second exemplary embodiment.
[0016] FIG. **10** is a perspective diagram illustrating a state where a third tray portion is additionally connected, according to a third exemplary embodiment.
[0017] FIG. **11** is a perspective diagram illustrating a state where a third tray portion is additionally added, according to a fourth exemplary embodiment.
[0018] FIG. **12** is a perspective diagram illustrating a state where a second grip portion is stored, according to the fourth exemplary embodiment.
DESCRIPTION OF THE EMBODIMENTS
[0019] Hereinbelow, exemplary embodiments of the present disclosure will be described with reference to the attached drawings.
Overall Configuration of Image Forming Apparatus

[0020] With reference to FIG. **1**, an overall configuration of a color image forming apparatus will be described. FIG. **1** is a longitudinal cross-section diagram illustrating a color laser printer A serving as an example of a color image forming apparatus. The color laser printer A serving also as an example of a storage apparatus is provided with a sheet feed cassette **100** for storing sheets. The color laser printer (image forming apparatus) A is provided with a feed roller **14** for feeding a sheet S stored in the sheet feed cassette **100** disposed at a bottom position of the color laser printer A, into the color laser printer A. The sheet feed cassette **100** is supported so as to be slidably movable with respect to the main body of the color laser printer A (hereinbelow, referred to as an apparatus main body). The sheet feed cassette **100** can be pulled out in an X direction with respect to the apparatus main body from a sheet mounted position where the stored sheet S is feedable by the feed roller **14**. The sheet feed cassette **100** can also be demounted from the apparatus main body.
[0021] At an upper right portion of the feed roller **14**, a registration roller unit **16** for adjusting the registration of the sheet S to convey the sheet S is disposed. Above the feed roller **14**, four process cartridges **1** (**1**Y, **1**M, **1**C, and **1**Bk) are disposed. Over the process cartridges **1** (**1**Y, **1**M, **1**C, and **1**Bk) including photosensitive drums **2** (**2**Y, **2**M, **2**C, **2**Bk), an intermediate transfer unit **70** is disposed so as to oppose the process cartridges **1** (**1**Y, **1**M, **1**C, and **1**Bk). A latent image on each of the photosensitive drums **2** is developed by a development unit (not illustrated). The intermediate transfer unit **70** includes an intermediate transfer belt **8**, and inside the intermediate transfer belt **8**, further includes primary transfer rollers **5** (**5**Y, **5**M, **5**C, and **5**Bk), a secondary transfer counter roller **10**, a tension roller **11**, and an intermediate transfer belt cleaning unit (not illustrated). The intermediate transfer belt **8** and the secondary transfer counter roller **10** form a transfer nip for nipping a sheet. On a right side of the intermediate transfer unit **70**, a secondary transfer roller **12** is disposed so as to oppose the secondary transfer counter roller **10**. The process cartridges **1**, the intermediate transfer unit **70**, the primary transfer rollers **5**, and the secondary transfer counter roller **10** constitute an image forming portion for forming an image on a sheet. Over a secondary transfer portion, a fixing unit **20** is disposed. Above a left portion of the fixing unit **20**, a sheet discharge roller pair **22**, a two-sided conveyance portion **24**, a reversing roller pair **23**, and a double-sided flapper **21** serving as a branching unit are disposed.
[0022] A toner image corresponding to desired image data is formed on each of the photosensitive drums **2**, and the toner image is primarily transferred from each of the photosensitive drums **2** onto the intermediate transfer belt **8**. On the other hand, the sheet S stored in the sheet feed cassette **100** is fed by the feed roller **14**. The sheet S fed by the feed roller **14** is conveyed by the registration roller unit **16** to the transfer nip. At the transfer nip, the toner image on the intermediate transfer belt **8** is transferred on the sheet S. The sheet S with the toner image transferred thereon is conveyed to the fixing unit **20**. The toner image transferred on the sheet is fixed onto the sheet by the fixing unit **20**. The sheet S with the toner image fixed thereon by the fixing unit **20** is discharged onto the stacking portion **90** by the sheet discharge roller pair **22**. In addition, in a case where images are to be formed on respective sides of the sheet, the sheet S is conveyed to the registration roller unit **16** again by the reversing roller pair **23**.
Configuration of Sheet Feed Cassette
[0023] With reference to FIGS. **2**, **3**, and **4**, the sheet feed cassette **100** will be described. FIG. **2** is a perspective diagram illustrating the sheet feed cassette **100**. FIG. **3** is a perspective diagram illustrating the sheet feed cassette **100** when it is slid to be extended. FIG. **4** is a perspective diagram illustrating a state where the sheet feed cassette **100** is separated.
[0024] The sheet feed cassette **100** is configured of two tray portions. More specifically, the sheet feed cassette **100** is configured of a first tray portion **60** on which a first grip portion **65** to be used when the sheet feed cassette **100** is pulled out from the front side of the apparatus main body is provided, and a second tray portion **50** located at the back side of the apparatus main body. On a front side of the main body of the first tray portion **60**, a cassette front cover **266** (see FIG. **4**) is provided. At a part of the cassette front cover **266**, the first grip portion **65** used when a user pulls

out the sheet feed cassette **100** is provided. More specifically, the first grip portion **65** provided on the cassette front cover **266** is exposed for the user to be able to grip it in a state where the sheet feed cassette **100** is located at its mounted position.

[0025] As illustrated in FIG. **4**, at respective right and left ends of the first tray portion **60**, pillar portions **61** and **62** are provided, and at a center portion of the first tray portion **60**, flange portions **63** and **64** are provided. On the second tray portion **50** serving as a movable member, a holding portion **51** including protrusions **51**A and **51**B at its upper portion, and a planar portion **51**C at its lower portion are provided, and engaged with the opposing pillar portion **61** of the first tray portion **60** in a slidable manner. Although not illustrated, a similar holding portion is provided on the opposite side of the second tray portion **50**, and engaged with another pillar portion **62** provided on the first tray portion **60** in a slidable manner. Further, at a center portion of the second tray portion **50**, a groove **55** is provided, and at an upper portion of the groove **55**, protrusions **53**A, **53**B, **54**A, and **54**B are provided. The flange portions **63** and **64**, provided on the first tray portion **60**, are insertable to the groove **55** and engaged by the groove **55** and the protrusions **53**A, **53**B, **54**A, and **54**B in a slidable manner. With the engagement at the right, left, and center, the first tray portion **60** and the second tray portion **50** are slidably engaged.

[0026] In a case where the second tray portion **50** is fully engaged with the first tray portion **60**, the sheet feed cassette **100** becomes the smallest size that can store A**4** size sheets (see FIG. **2**). Further, in a case where the second tray portion **50** is slid and moved with respect to the first tray portion **60** to increase the size of the sheet feed cassette **100**, the sheet feed cassette **100** becomes the maximum size that can store legal-size sheets (see FIG. **3**). In this way, the sheet feed cassette **100** is extendable and retractable in two sizes. Although not illustrated, stoppers for holding the positions at the minimum and maximum states, and a release lever for releasing the holding are provided, to allow a user to arbitrarily change the size of the sheet feed cassette **100**.

Third Tray Portion and Second Grip Portion

[0027] The sheet feed cassette **100** is normally configured of the first tray portion **60** and the second tray portion **50**.

[0028] However, in a case where a user desires to set sheets longer than the legal-size sheet, a third tray portion **110** can be connected to the sheet feed cassette **100** so that the third tray portion **110** is added to the sheet feed cassette **100**. By connecting the third tray portion **110**, the size of sheets that can be stored in the sheet feed cassette **100** can be extended to a larger size.

[0029] With reference to FIGS. **5** to **8**, a configuration of the third tray portion **110** serving as a connection unit will be described. FIG. **5** illustrates a state where the first tray portion **60**, the second tray portion **50**, and the third tray portion **110** are separated. FIG. **6** illustrates the third tray portion **110**. Further, FIG. **7** illustrates a state where the third tray portion **110** is additionally connected, and FIG. **8** illustrates a state where second grip portions **150** and **153** are open.

[0030] The third tray portion **110** is provided with stacking surfaces **111**A and **111**B for stacking the sheets S thereon, and pillar portions **112** and **113** are respectively provided on the right and left sides at an end portion of the third tray portion **110**. At a center portion of the third tray portion **110**, a recessed portion **120** is provided. On a side, on which the pillar portions **112** and **113** are provided, of the end portions of the recessed portion **120**, a flange portion **114** is provided. In a similar manner to the first tray portion **60**, the pillar portion **112** of the third tray portion **110** engages with the holding portion **51** provided on the second tray portion **50**. The pillar portion **113** on the opposite side engages with another holding portion (not illustrated) in a similar manner. The recessed portion **120** of the third tray portion **110** is insertable into the groove **55** provided in the second tray portion **50**. The flange portion **114** of the third tray portion **110** engages with the second tray portion **50** by the protrusions **53**B and **54**B of the second tray portion **50** in a similar manner to the first tray portion **60**. The second tray portion **50** is provided with pawl protrusions **255** and **56** and tap holes **57** and **58**. The third tray portion **110** is provided with screw holes **116** and **118** and engagement holes **115** and **117**. When the second tray portion **50** and the third tray portion **110** are engaged, the pawl protrusions **255** and **56** and the engagement holes **115** and **117** are respectively engaged, to match the positions of the screw holes **116** and **118** with the positions of the tap holes **57** and **58**, and the second tray portion **50** and the third tray portion **110** can be fixed with screws.

[0031] On the other hand, on the side opposite to the side engaged with the second tray portion **50**, holding portions **121** and **123** that form U shapes with the stacking surfaces **111**A and **111**B as lower surfaces are provided. Protrusions **122** (**122**A and **122**B) and **124** (**124**A and **124**B) are provided on the recessed portion **120**. The pillar portion **61** provided on the first tray portion **60** engages with the holding portion **121** of the third tray portion **110**, and the pillar portion **62** engages with the holding portion **123**, in a similar manner to the second tray portion **50**. The flange portions **63** and **64** provided on the first tray portion **60** are inserted into the recessed portion **120** of the third tray portion **110** to engage with each other by being pressed from above by the protrusions **122** and **124** of the third tray portion **110**, in a similar manner to the second tray portion **50**. Engaging holes **125** and **126** are provided in the stacking surfaces **111**A and **111**B of the third tray portion **110**, protrusions (not illustrated) are provided on the back side of the first tray portion **60**, and the engaging holes **125** and **126** engage with the protrusions. A screw hole **129** is provided in a side surface of the third tray portion **110**, and a screw tap hole **66** is provided in a side surface of the first tray portion **60**. When the first tray portion **60** and the third tray portion **110** engage with each other, the positions of the screw hole **129** and the screw tap hole **66** match to become able to be fixed with a screw. Although not illustrated, also the opposite side in the right-left direction of the sheet feed cassette **100** can be fixed with a screw.

[0032] A pair of the second grip portions **150** and **153** is provided approximately at a center portion of the third tray portion **110**. The second grip portion **153** is provided with a hand grip **154** formed of a metal wire rod for a user to grip. The hand grip **154** is pivotally supported to be rotatable by bearings **155**A and **155**B. A hand grip **151** formed of a metal wire rod pivotally supported to be rotatable by bearings **152**A and **152**B in a similar configuration is provided in the second grip portion **150** located on the side opposite to the second grip portion **153**. In the present exemplary embodiment, in a state where the sheet feed cassette **100** is located at a mounting position, the second grip portions **150** and **153** are covered by the apparatus main body, and a user cannot grip them.

[0033] The second grip portions **150** and **153** are respectively arranged on both sides of the sheet stacking area of the sheet feed cassette **100**. More specifically, the second grip portion **150** from among the second grip portions **150** and **153** is arranged on one side of the sheet stacking area of the sheet feed cassette **100** in a sheet width direction, and the second grip portion **153** is arranged on the other side of the sheet stacking area. In this case, the sheet width direction is a direction orthogonal to the sheet feed direction. In the present exemplary embodiment, in the sheet width direction, the second grip portions **150** and **153** are arranged with the center of gravity of the sheets stacked on the sheet feed cassette **100** being therebetween.

[0034] The hand grips **151** and **154** are painted blue different from colors of the components located around the hand grips **151** and **154** so as to be easily recognizable by a user. FIG. **8** illustrates a state where the second grip portions **150** and **153** are to be held.

[0035] The hand grips **151** and **154** rotationally move upward to be able to be gripped, and a user can grip and hold the sheet feed cassette **100**. Since the hand grips **151** and **154** are respectively arranged on external sides of the sheet storage portion (sheet stacking area), the sheet feed cassette **100** can be held even in a state where long sheets are fully stacked.

[0036] The hand grips **151** and **154** are arranged in such a manner that a user can grip them at a position above the sheet stacking surface of the sheet feed cassette **100**. In particular, in the present exemplary embodiment, the hand grips **151** and **154** are arranged in such a manner that a user can grip them at a position above the sheets stacked on the sheet feed cassette **100**.

[0037] The third tray portion **110** is arranged approximately at the center of the sheet feed cassette **100**, and the second grip portions **150** and **153** are arranged approximately at the center of the third tray portion **110**. Accordingly, since the second grip portions **150** and **153** are arranged approximately at the center of the sheet feed cassette **100**, to be near the center of gravity of the sheet feed cassette **100**, the sheet feed cassette **100** has a good weight balance and is easy to be held.

[0038] In addition, the third tray portion **110** is largest among the members constituting the sheet feed cassette **100**, is made of metal, and thus, is highest in rigidity. Since long and heavy long sheets are held by the large and rigid third tray portion **110** as the center, even in a state where the sheets are fully stacked, the deformation of the sheet feed cassette **100** is little and the sheet feed cassette **100** can support the sheets stably.

[0039] Because of the configuration that can support the sheets in the full stacked state as described above, when long sheets are newly set, the long sheets can be set in a state where the sheet feed cassette **100** is demounted from the apparatus main body, i.e., there is no obstacle on the sheet storage portion. Since the long sheets are long and heavy, and sometimes soft depending on the sheets, the sheets are difficult to hold, and the setting property is bad compared with the office standard-sized sheets such as A**4**. According to the present exemplary embodiment, it is possible to set the long sheets with no obstacle around the sheet feed cassette **100**, and even in a state where the long sheets are stacked, the sheet feed cassette **100** can be easily put back to the apparatus main body by holding the second grip portions **150** and **153**. Accordingly, the work is easily done.

[0040] If the second grip portions **150** and **153** are not provided, there are no positions for the user to hold the sheet feed cassette **100** after storing the long sheets in the sheet feed cassette **100**, and thus, the user has to once lift the sheet feed cassette **100** a little and place the user's hand under the bottom of the sheet feed cassette **100** to hold the sheet feed cassette **100**. In this case, many actions are necessary, and since the user has to put the user's hand to an out of sight position, there is a risk that the user supports a functional portion or a weak portion in strength, which may impair the function and may be undesirable.

Other Exemplary Embodiments

[0041] Modification examples of the second grip portions will be described. FIG. **9** is a diagram illustrating a second exemplary embodiment. A pair of second grip portions **171** and **174** is respectively provided approximately at center portions of a third tray portion **170**. The second grip portions **171** and **174** are respectively provided with hand grips **172** and **175** formed by externally bending and protruding from side surfaces of the third tray portion **170**, and a user can grip the hand grips **172** and **175**. In order to inform the user clearly that the hand grips **172** and **175** are holding portions, blue labels **173** and **176** different in color from the components around them are respectively attached. In the exemplary embodiment in FIG. **9**, the labels **173** and **176** are solid color labels, but characters or illustrations may be added thereto. Since the hand grips **172** and **175** are respectively arranged on external portions of the sheet storage portion, the user can hold the sheet feed cassette **100** with the long sheets fully stored.

[0042] FIG. **10** is a diagram illustrating a third exemplary embodiment. A pair of second grip portions **181** and **185** is respectively provided approximately at center portions of a third tray portion **180**. On the upper sides of walls **182** and **186** of the sheet storage portion, grip holes **183** and **187** and hand grips **184** and **188** are respectively provided. Since the grip holes **183** and **187** and the hand grips **184** and **188** are provided at positions higher than the top sheet surface of the fully stacked long sheets, a user can hold the sheet feed cassette **100** with the long sheets fully stored.

[0043] FIGS. **11** and **12** are diagrams illustrating a fourth exemplary embodiment. A third tray portion **190** is not provided with the second grip portion, and a second tray portion **200** is provided with one second grip portion **201**. FIG. **11** illustrates a state where the second grip portion **201** is open to be holdable, and FIG. **12** illustrates a state where the second grip portion **201** is folded. The second grip portion **201** is provided on a side surface **210** of the second tray portion **200**. The second grip portion **201** includes a hand grip **202** and a spring **203**. The hand grip **202** is provided with a pair of rotation shafts **202***a* and **202***b*, and is pivotally supported to be rotatable by the side surface **210** of the second tray portion **200**. The rotation shaft **202***b* is provided with the spring **203** and urged toward a direction in which the hand grip **202** opens. The hand grip **202** is formed in blue different from colors of the components around the hand grip **202**. The hand grip **202** is prevented from opening, by a member in the apparatus main body when the sheet feed cassette **100** is mounted in the apparatus main body. When the sheet feed cassette **100** is pulled out from the apparatus main body, the hand grip **202** is opened by a spring urging force, and a user can hold the sheet feed cassette **100** by the user's hand placing on the hand grip **202**. In addition, only one second grip portion **201** is provided on the sheet feed cassette **100**, but when a user holds the sheet feed cassette **100**, the user can hold the sheet feed cassette **100** by holding the first grip portion **65** with one hand and holding the second grip portion **201** with the other hand. The first grip portion **65** and the second grip portion **201** are desirably arranged diagonally as much as possible, and in the present exemplary embodiment, the first grip portion **65** is arranged on the left side surface on the front side of the sheet feed cassette **100**, and the second grip portion **201** is arranged on the right side surface on the back side of the sheet feed cassette **100**, in consideration of the balance when the user holds the sheet feed cassette **100**. With this configuration, it is possible for a user to stably hold the sheet feed cassette **100** with the long sheets fully stacked and put the sheet feed cassette **100** back to the apparatus main body.

[0044] With any of the exemplary embodiments described above, in a state where the long sheets are stored, a user can stably hold the heavy sheet feed cassette **100** with the long sheets fully stacked thereon by providing two or more grip portions for holding the sheet feed cassette **100**. Accordingly, the workability of setting the sheet feed cassette **100** into the apparatus main body is improved. Further, the workability of setting again the sheet feed cassette **100** demounted from the apparatus main body of the color laser printer (color image forming apparatus) A thereinto is improved. Thus, it is possible to recommend the work of setting the long sheets to the sheet feed cassette **100** in a state where the sheet feed cassette **100** is demounted from the color laser printer A. In addition, since the long and heavy long sheets difficult to hold can be set (stored) in the sheet feed cassette **100** in a wide space, the workability is good in this point. As the workability is good, when the long sheets are stored into the sheet feed cassette **100**, the risk of damaging the long sheets can be easily avoided.

[0045] In the exemplary embodiments described above, the electrophotographic image forming unit (color laser printer A) is exemplified, but the image forming unit may be an ink-jet method image forming unit.

[0046] The long sheet is heavier than sheets other than the long sheet. In a case where a user holds a sheet feed cassette with the long sheets stored therein in a state where the sheet feed cassette is demounted from the apparatus main body, when a user holds a weak portion in strength, there is a risk of damaging the sheet feed cassette. If a grip portion or grip portions are not provided on the sheet feed cassette, a user may touch the functionally important portion by mistake. In such a case, there may be a risk of impairing the function of the sheet feed cassette. In addition, it may be considered that a user sets the long sheets without demounting the sheet feed cassette from the apparatus main body, but in this case, since the sheets cannot be set in a manner to place the sheets down in the sheet feed cassette from directly above, the workability is not good and there is a risk of damaging the sheets. According to the exemplary embodiments described above, as the sheet feed cassette is provided with a grip portion or grip portions, the workability can be improved.

[0047] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0048] This application claims the benefit of Japanese Patent Application No. 2024-037566, filed Mar. 11, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A storage apparatus comprising: an apparatus main body; and a cassette for storing a sheet to be fed, the cassette being configured to be pulled out from the apparatus main body and demounted from the apparatus main body, wherein the cassette includes: a first tray portion configured to support the sheet, a second tray portion configured to support the sheet, a connection portion configured to connect the first tray portion and the second tray portion, a first grip portion used when the cassette is pulled out from the apparatus main body, and a second grip portion provided on the connection portion, and used when the cassette is held in a state where the cassette is demounted from the apparatus main body.
2. The storage apparatus according to claim 1, wherein the first tray portion and the second tray portion are configured to be connectable not via the connection portion, and wherein a size of the sheet storable in the cassette is larger in a case where the first tray portion and the second tray portion are connected via the connection portion, than that in a case where the first tray portion and the second tray portion are connected not via the connection portion.
3. The storage apparatus according to claim 1, further comprising a plurality of second grip portions including the second grip portion, wherein the plurality of second grip portions is provided so that the cassette can be held in a state where only the plurality of second grip portions is held.
4. The storage apparatus according to claim 1, wherein the second grip portion is arranged on each side of a sheet stacking area of the cassette.
5. The storage apparatus according to claim 1, further comprising at least two second grip portions including the second grip portion, and wherein one of the at least two second grip portions is arranged on one side of a sheet stacking area of the cassette, and another one of the at least two second grip portions is arranged on another side of the sheet stacking area of the cassette, in a sheet width direction orthogonal to a feeding direction in which the sheet is fed.
6. The storage apparatus according to claim 1, wherein the second grip portion is arranged above a sheet stacking surface of the cassette so that a user can place a user's hand thereon.
7. The storage apparatus according to claim 1, wherein the second grip portion is arranged above the sheet stacked in the cassette so that a user can place a user's hand thereon.
8. The storage apparatus according to claim 1, wherein the first grip portion is exposed, and the second grip portion is covered by the apparatus main body so as not to be gripped, in a state where the cassette is located at a mounted position at which the stored sheet is feedable by a sheet feed unit.
9. A cassette for storing a sheet to be fed, comprising: a first tray portion configured to support the sheet; a second tray portion configured to support the sheet; a connection portion configured to connect the first tray portion and the second tray portion; a first grip portion used when the cassette is pulled out from an apparatus main body; and a second grip portion provided on the connection portion and used when a user holds the cassette.
10. The cassette according to claim 9, wherein the first tray portion and the second tray portion are configured to be connectable not via the connection portion, and wherein a size of the sheet storable in the cassette is larger in a case where the first tray portion and the second tray portion are connected via the connection portion, than that in a case where the first tray portion and the second tray portion are connected not via the connection portion.
11. The cassette according to claim 9, further comprising a plurality of second grip portions including the second grip portion, wherein the plurality of second grip portions is provided so that the cassette can be held in a state where only the plurality of second grip portions is held.
12. The cassette according to claim 9, wherein the second grip portion is arranged on each side of a sheet stacking area of the cassette.
13. The cassette according to claim 9, further comprising at least two second grip portions

including the second grip portion, wherein one of the at least two second grip portions is arranged on one side of a sheet stacking area in the cassette in a sheet width direction orthogonal to a sheet feeding direction in which the sheet is fed, and wherein another one of the at least two second grip portions is arranged on another side of the sheet stacking area.

14. The cassette according to claim 9, wherein the second grip portion is arranged above a sheet stacking surface of the cassette so that a user can place a user's hand thereon.

15. The cassette according to claim 9, wherein the second grip portion is arranged above the sheet stacked in the cassette so that a user can place a user's hand thereon.

16. The cassette according to claim 9, wherein the first grip portion is exposed, and the second grip portion is covered by the apparatus main body so as not to be gripped, in a state where the cassette is located at a mounted position at which the stored sheet is feedable by a sheet feed unit.

17. An image forming apparatus comprising: an apparatus main body including an image forming unit; a cassette for storing a sheet, and configured to be pulled out and demounted from the apparatus main body; and a sheet feed unit configured to feed the sheet stored in the cassette to the image forming unit, wherein the cassette includes: a first tray portion configured to support the sheet, a second tray portion configured to support the sheet, a connection portion configured to connect the first tray portion and the second tray portion, a first grip portion used when the cassette is pulled out from the apparatus main body, and a second grip portion provided on the connection portion, and used when the cassette is held in a state where the cassette is demounted from the apparatus main body.